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3D AoA Calibration procedure Rev B

Dec, 2022

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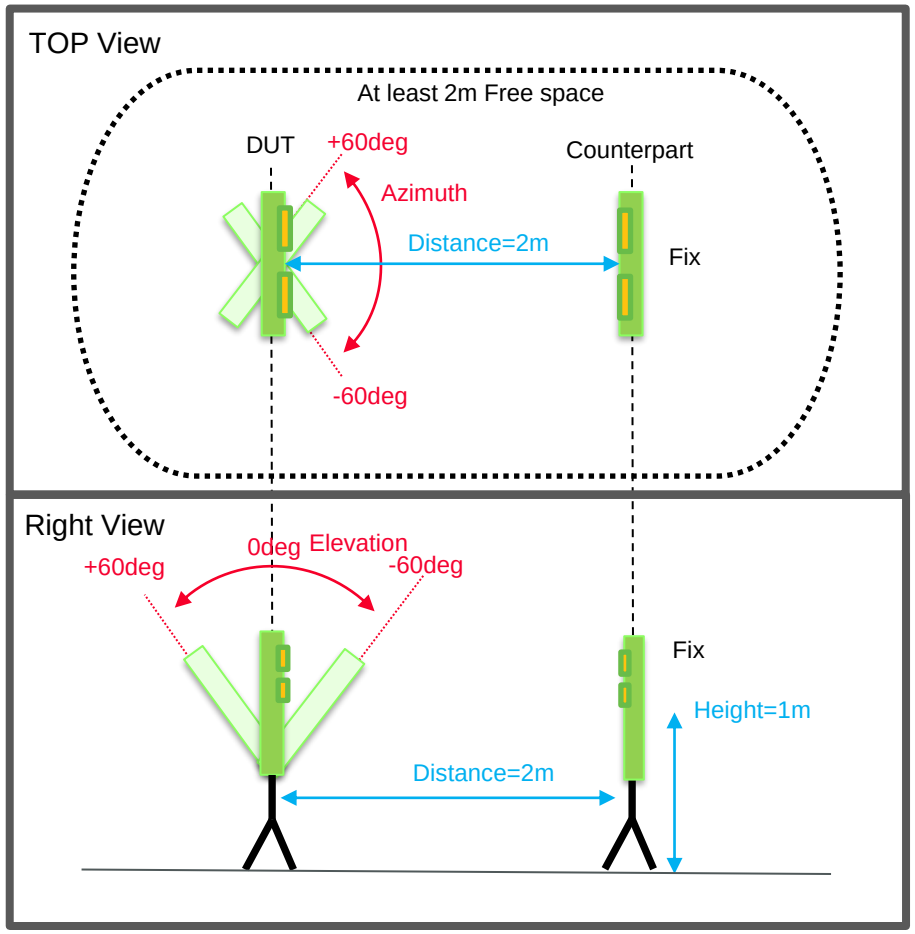
Setup of the measurement environment



Find a free space of at least 2m around DUT and counterpart. (6m x 4m at minimum)

Recommend DUT setup as 1m height and 2m distance.

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SW setup for measurement

Software Guide

Title
Quick Start Guide
Starting Software Development Guide

Write the FW for PnP mode to the product and the counterpart (recommend using Type2BP EVK).

For FW for PnP, use the SDK sample program.

For information on how to build FW, see the ” Build Sample Application(pnp) in Type 2BP Starting Software Development Guide”.

Test Guide

Title
AoA Test Result
PnP Python Script for SDK v03.13.03
PnP Python Script for SDK v03.14.05/v03.15.11
PnP Test Guide

To run PnP FW, use a Python script on the PC side.
Note that the script files used vary for each version of the SDK.

See the “PnP Test Guide” for how PnP mode work.

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SW setup for measurement

Advance preparation: (**using SDK v03.15.11**)

Use the “2bp_pnp_v03.15.11.zip” from the “Type2BP SDK UWBIOT SR150 v03.15.11 MCUx” site.

Write “2bp_pnp_v03.15.11.bin” on the product.

Type2BP SDK UWBIOT SR150 v03.15.11 MCUx

Title	Date	File name
UWBIOT SR150 v03.15.11 MCUx	2022/7/1	UWBIOT_SR150_v03.15.11_MCUx.zip *Important* If you use a version of the EVK prior to Rev3.4, you will need to modify the EVK. See document(MTD-APN-029-A_Type2BP_EVK_rework_guide_for_Rev3.X.pdf) for more details. This document is available on Type2BP Document site.
PnP binary for Type2BP EVK (v03.15.11)	2022/7/20	2bp_pnp_v03.15.11.zip

Python script file used by PnP will be “MTD-SCP-026-A_DS-TWR_SR150_Unicast_v03.14.05.py”.

For information on how to operate in PnP mode,

See “MTD-APN-007-E_Type2BP_EVK_PnP_binary_test_guide.pdf”.

Test Guide

Title	Date	File name
AoA Test Result	2021/11/11	MTD-TDR-005-A_Type2BP_EVK_3D_AoA_test_result.pdf
PnP Python Script for SDK v03.13.03	2022/4/4	MTD-SCP-025-A_DS-TWR_SR150_Unicast_FW32.zip
PnP Python Script for SDK v03.14.05/v03.15.11	2022/6/28	MTD-SCP-026-A_DS-TWR_SR150_Unicast_v03.14.05.py
PnP Test Guide	2022/6/22	MTD-APN-007-E_Type2BP_EVK_PnP_binary_test_guide.pdf

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SW setup for measurement

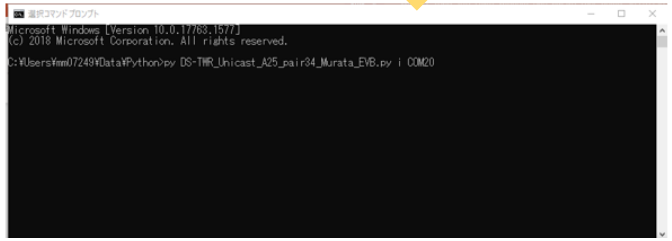
Operation image:

Initiator

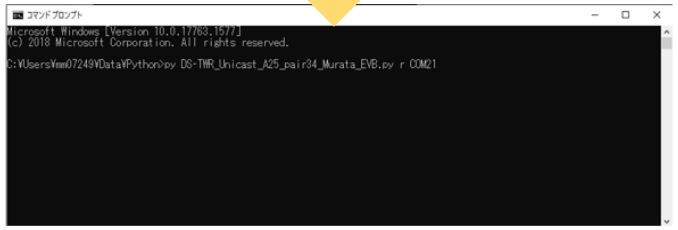
Responder

Flashed pnp binary file

Put the Python script file



Run the Python script file
in Command prompt



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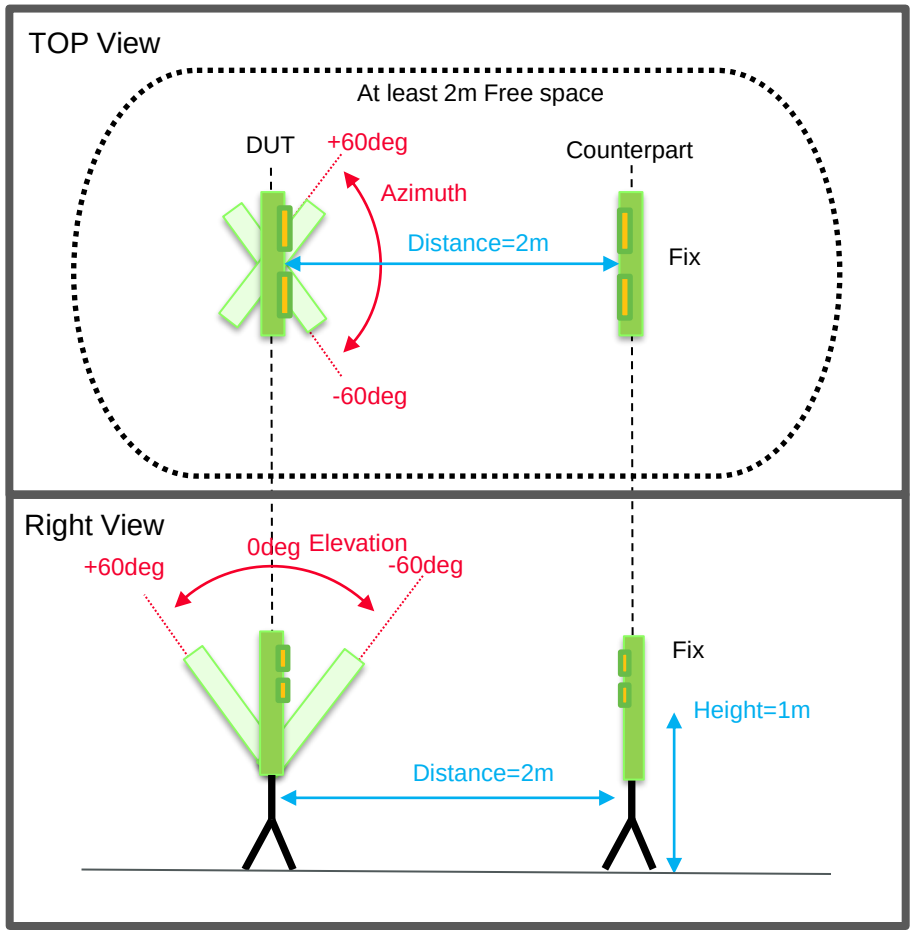
Acquisition of measurement data

3D AoA case:



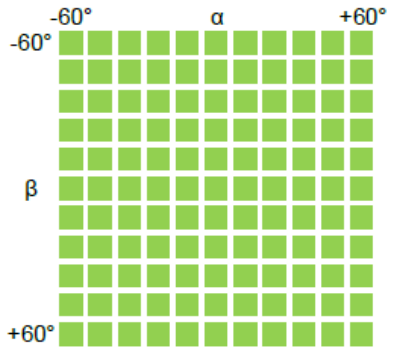
Obtain data in PnP mode.

Each Azimuth / Elevation 12deg step

Azimuth: -60 ~ +60 deg

Elevation: -60 ~ +60 deg

Measure more than 10times average



Measuring point

MTD-SCP-026-A_DS-TWR_SR150_Unicast_v03.14.05.py
Please check the “UWB_SESSION_SET_APP_CONFIG_NXP”.

```
    # Proprietary
UWB_SESSION_SET_APP_CONFIG_NXP = [0x21, 0x03, 0x00, 0x31] + SESSION_ID + [
    0x0A,                                       # Number of parameters
    0xE3, 0x01, 0x01, 0x76,                     # CIR_CAPTURE_MODE
####0xE3, 0x02, 0x01, 0x01,                     # MAC_PAYLOAD_ENCRYPTION
####0xE3, 0x03, 0x01, 0x01,                     # RX_ANTENNA_POLARIZATION_OPTION
    0xE3, 0x05, 0x01, 0x03,                     # SESSION_SYNC_ATTEMPTS
    0xE3, 0x06, 0x01, 0x03,                     # SESSION_SHED_ATTEMPTS
    0xE3, 0x07, 0x01, 0x00,                     # SCHED_STATUS_NTF
    0xE3, 0x08, 0x01, 0x00,                     # TX_POWER_DELTA_FCC
    0xE3, 0x09, 0x01, 0x00,                     # TEST_KDF_FEATURE
    0xE3, 0x0B, 0x01, 0x00,                     # TX_POWER_TEMP_COMPENSATION
####0xE3, 0x0C, 0x01, 0x03,                     # WIFI_COEX_MAX_TOLERANCE_COUNT
####0xE3, 0x0D, 0x01, 0x00,                     # ADAPTIVE_HOPPING_THRESHOLD
####0xE3, 0x13, 0x01, 0x00,                     # AUTHENTICITY_TAG
####0xE3, 0x14, 0x02, 0x1E, 0x14,               # RX_NBIC_CONFIG
    0xE3, 0x15, 0x01, 0x03,                     # MAC_CFG
####0xE3, 0x16, 0x01, 0x00                      # SESSION_INBAND_DATA_TX_BLOCKS
####0xE3, 0x17, 0x01, 0x00                      # SESSION_INBAND_DATA_RX_BLOCKS
####0xE3, 0x18, 0x01, 0x00                      # SUSPEND_RANGING
####0xE3, 0x19, 0x01, 0x00                      # RX_ANTENNA_SELECTION_RFM
####0xE3, 0x1A, 0x01, 0x00                      # DATA_TRANSFER_MODE
    0xE3, 0x1B, 0x02, 0x01, 0x00                # ANTENNAS_CONFIGURATION_TX, supported from FW32
    0xE3, 0x1C, 0x04, 0x00, 0x02, 0x01, 0x02    # ANTENNAS_CONFIGURATION_RX, supported from FW32, for 3D AoA
]
```

Please change the setting for 3D AoA mode before the calibration.

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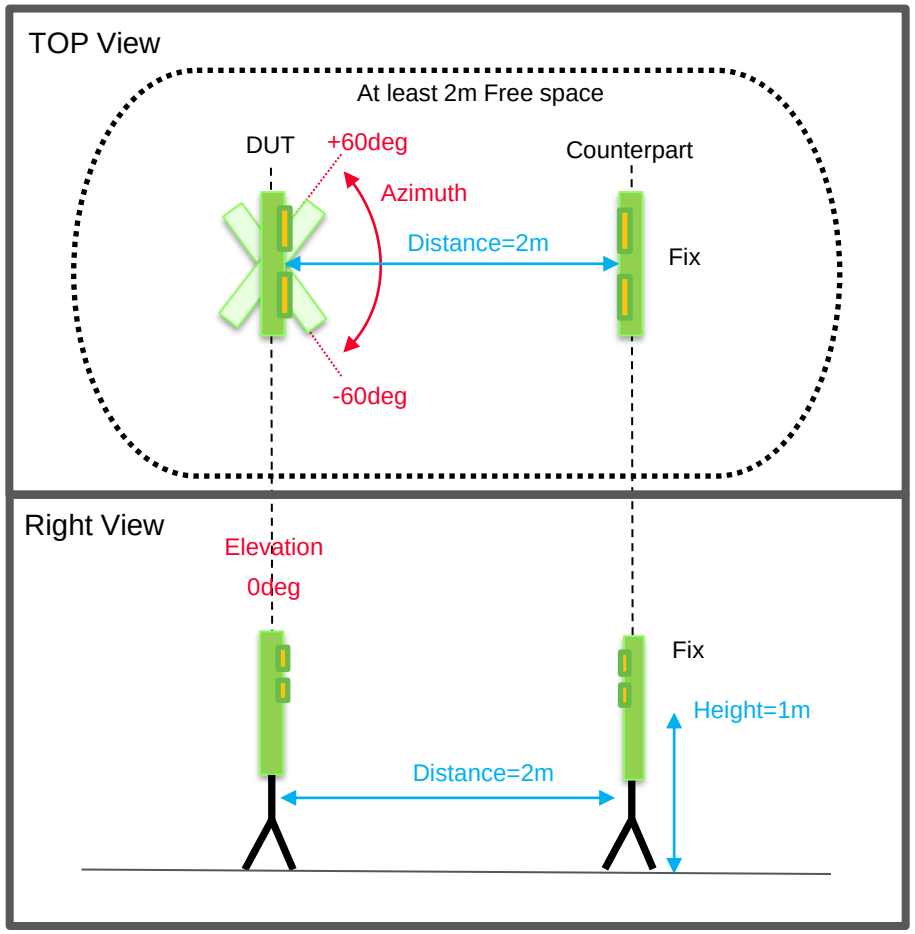
Acquisition of measurement data

2D AoA case:



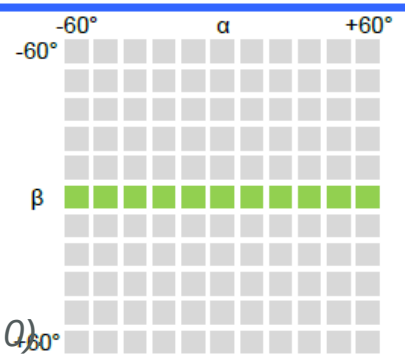
Obtain data in PnP mode.

Each Azimuth 12deg step

Azimuth: -60 ~ +60 deg

Measure more than 10times average

When using 2D AoA, only Azimuth is measured(Elevation is 0).



Measuring point

MTD-SCP-026-A_DS-TWR_SR150_Unicast_v03.14.05.py
Please check the “UWB_SESSION_SET_APP_CONFIG_NXP”.

```
    # Proprietary
UWB_SESSION_SET_APP_CONFIG_NXP = [0x21, 0x03, 0x00, 0x31] + SESSION_ID + [
    0x0A,                                       # Number of parameters
    0xE3, 0x01, 0x01, 0x76,                     # CIR_CAPTURE_MODE
####0xE3, 0x02, 0x01, 0x01,                     # MAC_PAYLOAD_ENCRYPTION
####0xE3, 0x03, 0x01, 0x01,                     # RX_ANTENNA_POLARIZATION_OPTION
    0xE3, 0x05, 0x01, 0x03,                     # SESSION_SYNC_ATTEMPTS
    0xE3, 0x06, 0x01, 0x03,                     # SESSION_SHED_ATTEMPTS
    0xE3, 0x07, 0x01, 0x00,                     # SCHED_STATUS_NTF
    0xE3, 0x08, 0x01, 0x00,                     # TX_POWER_DELTA_FCC
    0xE3, 0x09, 0x01, 0x00,                     # TEST_KDF_FEATURE
    0xE3, 0x0B, 0x01, 0x00,                     # TX_POWER_TEMP_COMPENSATION
####0xE3, 0x0C, 0x01, 0x03,                     # WIFI_COEX_MAX_TOLERANCE_COUNT
####0xE3, 0x0D, 0x01, 0x00,                     # ADAPTIVE_HOPPING_THRESHOLD
####0xE3, 0x13, 0x01, 0x00,                     # AUTHENTICITY_TAG
####0xE3, 0x14, 0x02, 0x1E, 0x14,               # RX_NBIC_CONFIG
    0xE3, 0x15, 0x01, 0x03,                     # MAC_CFG
####0xE3, 0x16, 0x01, 0x00                      # SESSION_INBAND_DATA_TX_BLOCKS
####0xE3, 0x17, 0x01, 0x00                      # SESSION_INBAND_DATA_RX_BLOCKS
####0xE3, 0x18, 0x01, 0x00                      # SUSPEND_RANGING
####0xE3, 0x19, 0x01, 0x00                      # RX_ANTENNA_SELECTION_RFM
####0xE3, 0x1A, 0x01, 0x00                      # DATA_TRANSFER_MODE
    0xE3, 0x1B, 0x02, 0x01, 0x00,               # ANTENNAS_CONFIGURATION_TX, supported from FW32
#   0xE3, 0x1C, 0x04, 0x00, 0x02, 0x01, 0x02    # ANTENNAS_CONFIGURATION_RX, supported from FW32, for 2D AoA
    0xE3, 0x1C, 0x04, 0x00, 0x02, 0x01, 0x00    # ANTENNAS_CONFIGURATION_RX, supported from FW32, for 2D AoA
]
```

Please change the setting for 2D AoA mode before the calibration.

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Check the measurement data

Collect PDoA1 and PDoA2 data with PnP mode.

Operation log data in PnP mode.

```
***(168) NLos:0   Dist:128   Azimuth:36.000000 (FOM:100)   Elevation:60.000000 (FOM:100)  PDoA1:-159.031250
PDoA2:156.125000
*** Avg Dist:121   Avg Azimuth:-12.500000   Avg Elevation:26.900000   Avg_PDoA1:-1.987500   Avg_PDoA2:60.154688
2022-11-04 16:51:04.141NXPUCIR <= 62 00 00 50 a9 00 00 00 01 00 00 00 00 c8 00 00 00 01 00 00 00 00 00 00 00 00 00
00 01 22 22 00 00 7e 00 00 12 64 00 1e 64 00 00 00 00 00 00 05 00 00 00 00 00 00 00 00 00 00 00 00 16 00 0f aa b4 2b b4
00 00 02 68 b2 41 03 46 44 41 03 00 01 02 00 00 00
***(169) NLos:0   Dist:126   Azimuth:36.000000 (FOM:100)   Elevation:60.000000 (FOM:100)  PDoA1:-155.187500
PDoA2:136.546875
*** Avg Dist:121   Avg Azimuth:-4.670000   Avg Elevation:33.920000   Avg_PDoA1:-26.145312   Avg_PDoA2:75.832813
2022-11-04 16:51:04.348NXPUCIR <= 62 00 00 50 aa 00 00 00 01 00 00 00 00 c8 00 00 00 01 00 00 00 00 00 00 00 00 00
00 01 22 22 00 00 7d 00 00 12 64 00 1e 64 00 00 00 00 00 00 05 00 00 00 00 00 00 00 00 00 00 00 00 16 00 0f ac b5 4f b4
00 00 02 88 b3 41 03 82 49 41 03 00 01 02 00 00 00
***(170) NLos:0   Dist:125   Azimuth:36.000000 (FOM:100)   Elevation:60.000000 (FOM:100)  PDoA1:-152.937500
PDoA2:147.015625
```

MTD-SCP-026-A_DS-TWR_SR150_Unicast_v03.14.05.py already has AoA calibration parameter for Murata EVK.
If customer use this script on their product, the Azimuth (AoA1), Elevation(AoA2) result should be far from real angle because antenna performance should be different from Murata EVK. (this is normal)
This is normal, please ignore AoA1, AoA2 result on calibration phase, and please collect PDOA1 and PDOA2 data.
*If 2D AoA case, Elevation and PDoA2 shows 0.0.

After NXP generates calibration parameter for your product, AoA1, AoA2 result should close to real angle.

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Collect the measurement data

Please enter the data obtained in PnP mode in ”Example_data_for_AoA_CAL_20211026.xlsx” file.

```
***(168) NLos:0   Dist:128   Azimuth:36.000000 (FOM:100)   Elevation:60.000000 (FOM:100)  PDoA1:-159.031250
PDoA2:156.125000
*** Avg Dist:121   Avg Azimuth:-12.500000   Avg Elevation:26.900000   Avg_PDoA1:-1.987500   Avg_PDoA2:60.154688
2022-11-04 16:51:04.141NXPUCIR <= 62 00 00 50 a9 00 00 00 01 00 00 00 00 c8 00 00 00 01 00 00 00 00 00 00 00 00 00
00 01 22 22 00 00 7e 00 00 12 64 00 1e 64 00 00 00 00 00 00 05 00 00 00 00 00 00 00 00 00 00 00 00 00 16 00 0f aa b4 2b b4
00 00 02 68 b2 41 03 46 44 41 03 00 01 02 00 00 00
***(169) NLos:0   Dist:126   Azimuth:36.000000 (FOM:100)   Elevation:60.000000 (FOM:100)  PDoA1:-155.187500
PDoA2:136.546875
*** Avg Dist:121   Avg Azimuth:-4.670000   Avg Elevation:33.920000   Avg_PDoA1:-26.145312   Avg_PDoA2:75.832813
2022-11-04 16:51:04.348NXPUCIR <= 62 00 00 50 aa 00 00 00 01 00 00 00 00 c8 00 00 00 01 00 00 00 00 00 00 00 00 00
00 01 22 22 00 00 7d 00 00 12 64 00 1e 64 00 00 00 00 00 00 05 00 00 00 00 00 00 00 00 00 00 00 00 00 16 00 0f ac b5 4f b4
00 00 02 88 b2 41 03 82 49 41 03 00 01 02 00 00 00
***(170) NLos:0   Dist:125   Azimuth:36.000000 (FOM:100)   Elevation:60.000000 (FOM:100)  PDoA1:-152.937500
PDoA2:147.015625
```

Console log

Please fill in to below blank.

DUT name		FW ver		used Python script	DS-TWR_xxxxxxxxx.py
Channel		Anntena pitch		XX mm	

Please share us raw data of more than 10 times measurement data, not only Average data.

Example data for 30times measurement.

H	V			NLOS	Dist	AoA1	AoA2	PDoA1	PDoA2
						Azimuth	Elevation		
-60	60	***(0)	NLos	0	191	-60	-60	133.5781	-63.2813
-60	-60	***(1)	NLos	0	188	-60	-60	134.7813	-54.8438
-60	-60	***(2)	NLos	0	190	-60	-60	133.4844	-63.0469
-60	-60	***(3)	NLos	0	187	-60	-60	127.625	-61.9063
-60	-60	***(4)	NLos	0	183	-60	-60	132.8594	-65.1094
-60	-60	***(5)	NLos	0	189	-60	-60	132.1875	-54.3281
-60	-60	***(6)	NLos	0	193	-60	-60	131.7344	-62.0938
-60	-60	***(7)	NLos	0	184	-60	-60	126.0313	-57.1875
-60	-60	***(8)	NLos	0	186	-60	-60	126.8906	-61.5156
-60	-60	***(9)	NLos	0	182	-60	-60	131.9375	-60.0938

Example_data_for_AoA_CAL_20211026.xlsx

The Excel format is to calculate the average value of 30 measurements.
If you change the measurement conditions, please modify the format.

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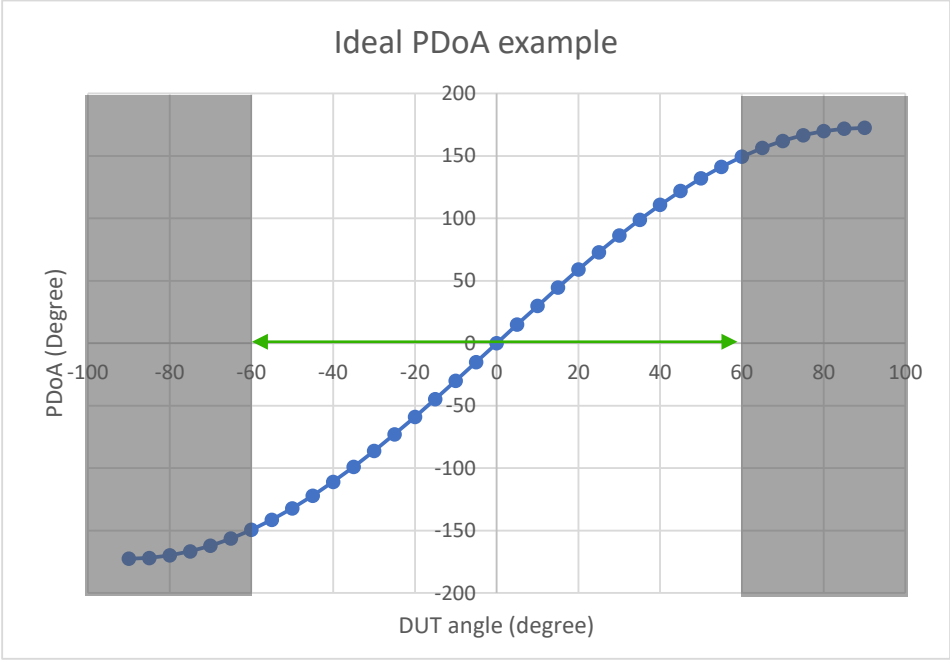
Check the measurement data

Please check the measurement data.

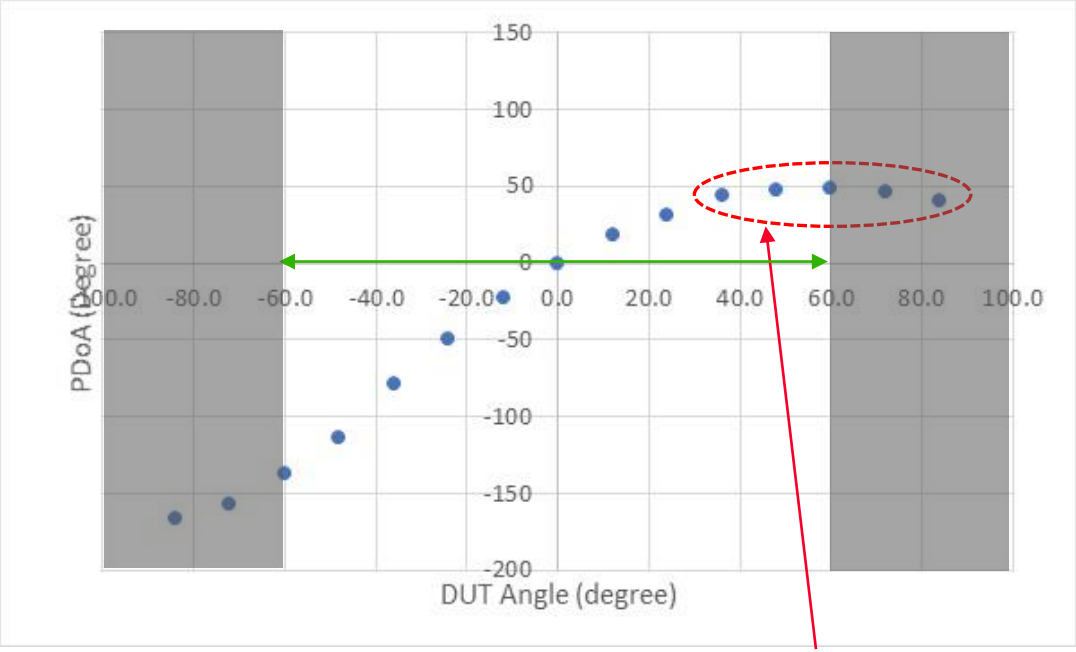
PDoA result must be stable and monotonic.

Monotonic case example



This is ideal PDoA @ 8GHz.

Non monotonic case example



With this PDoA performance, AoA result around DUT angle 40degree~ will not be reliable. Please reconfirm the measurement environment or antenna placement.

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Feed back the data to Murata

Please e-mail us with following format.

E-mail address : uwb-nxp-module@murata.com

Subject : Request for 3D-AoA Calibration data for 2BP
Body :
Your company name :
Your name :
Attachment :
Example_data_for_AoA_CAL_20211026.xlsx

We will contact you as soon as we know the result.
Murata will provide updated information with calibration parameters.

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Verify the updated calibration parameters

Verify AoA1(Azimuth) and AoA2(Elevation) performance with new script.

Operation log data in PnP mode.

```
***(168) NLos:0   Dist:128   Azimuth:36.000000 (FOM:100)   Elevation:60.000000 (FOM:100)  PDoA1:-159.031250
PDoA2:156.125000
*** Avg Dist:121   Avg Azimuth:-12.500000   Avg Elevation:26.900000   Avg_PDoA1:-1.987500   Avg_PDoA2:60.154688
2022-11-04 16:51:04.141NXPUCIR <= 62 00 00 50 a9 00 00 00 01 00 00 00 00 c8 00 00 00 01 00 00 00 00 00 00 00 00 00
00 01 22 22 00 00 7e 00 00 12 64 00 1e 64 00 00 00 00 00 00 05 00 00 00 00 00 00 00 00 00 00 00 00 16 00 0f aa b4 2b b4
00 00 02 68 b2 41 03 46 44 41 03 00 01 02 00 00 00
***(169) NLos:0   Dist:126   Azimuth:36.000000 (FOM:100)   Elevation:60.000000 (FOM:100)  PDoA1:-155.187500
PDoA2:136.546875
*** Avg Dist:121   Avg Azimuth:-4.670000   Avg Elevation:33.920000   Avg_PDoA1:-26.145312   Avg_PDoA2:75.832813
2022-11-04 16:51:04.348NXPUCIR <= 62 00 00 50 aa 00 00 00 01 00 00 00 00 c8 00 00 00 01 00 00 00 00 00 00 00 00 00
00 01 22 22 00 00 7d 00 00 12 64 00 1e 64 00 00 00 00 00 00 05 00 00 00 00 00 00 00 00 00 00 00 00 16 00 0f ac b5 4f b4
00 00 02 88 b3 41 03 82 49 41 03 00 01 02 00 00 00
***(170) NLos:0   Dist:125   Azimuth:36.000000 (FOM:100)   Elevation:60.000000 (FOM:100)  PDoA1:-152.937500
PDoA2:147.015625
```

Please confirm the AoA1 and AoA2 result that should close to real angle.

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3D AoA Calibration example data

Nov, 2021

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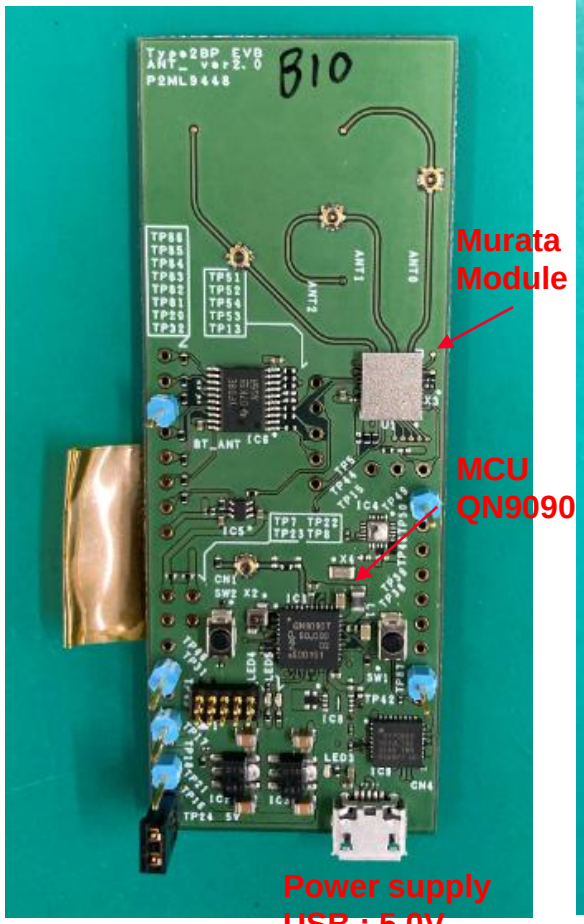
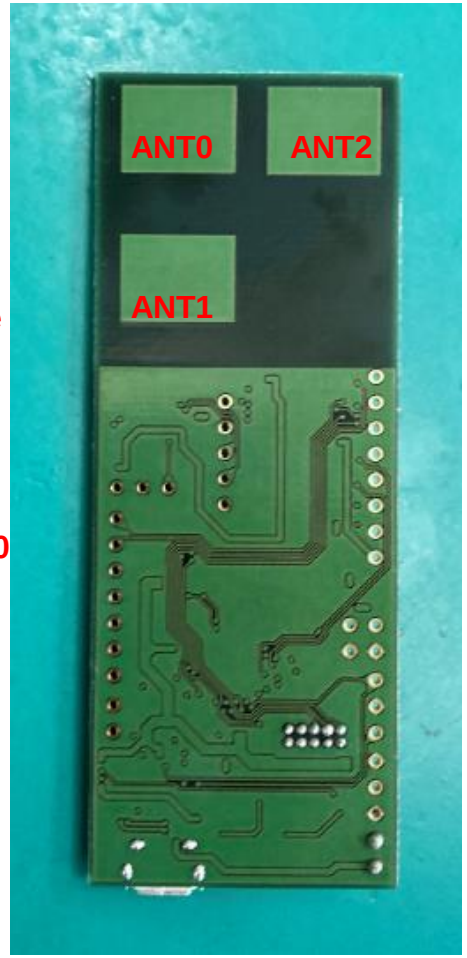
DUT : Murata module EVB



<TOP>



<Bottom>

Murata module : ES2.1
EVB : Ver2.0

SDK : UWB01_SW_FreeRTOS_RHD_A27
ROW FW : HELIOS_B1_FW_PROD_27.00.bin
PnP mode
Script : DS-TWR_Unicast_A25_pair34_Murata_EVB.py.txt

Initiator : EVB# B10
Responder : EVB# B9

Location : Murata HQ meeting room

Tilt : -60deg to +60deg : 12deg step
Azimuth : -60deg to +60deg : 12deg step
Average : 30 times

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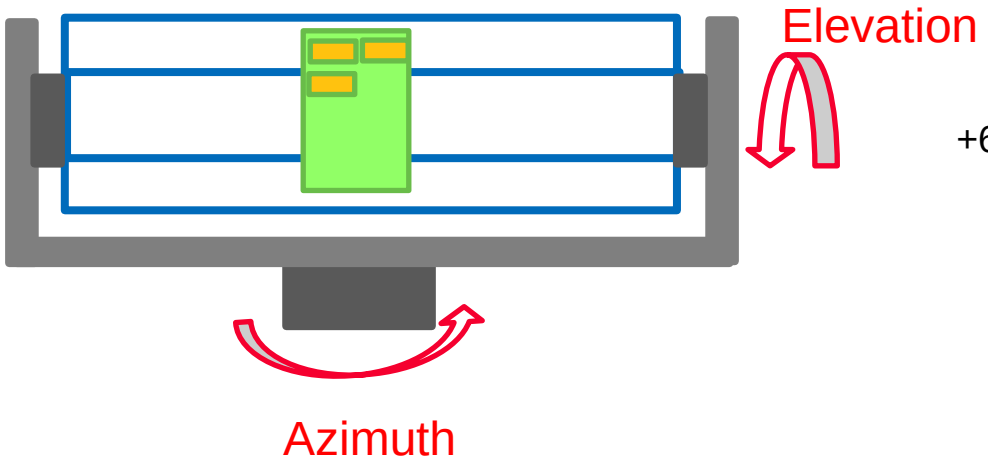
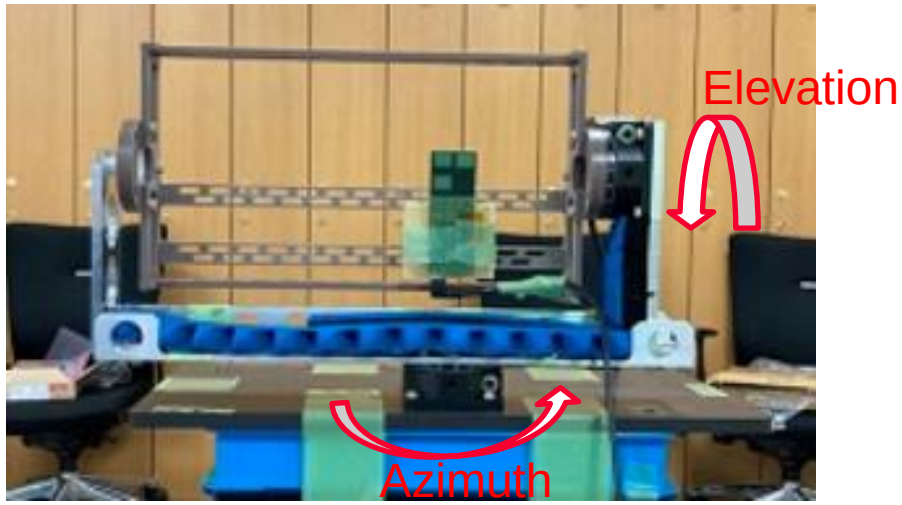
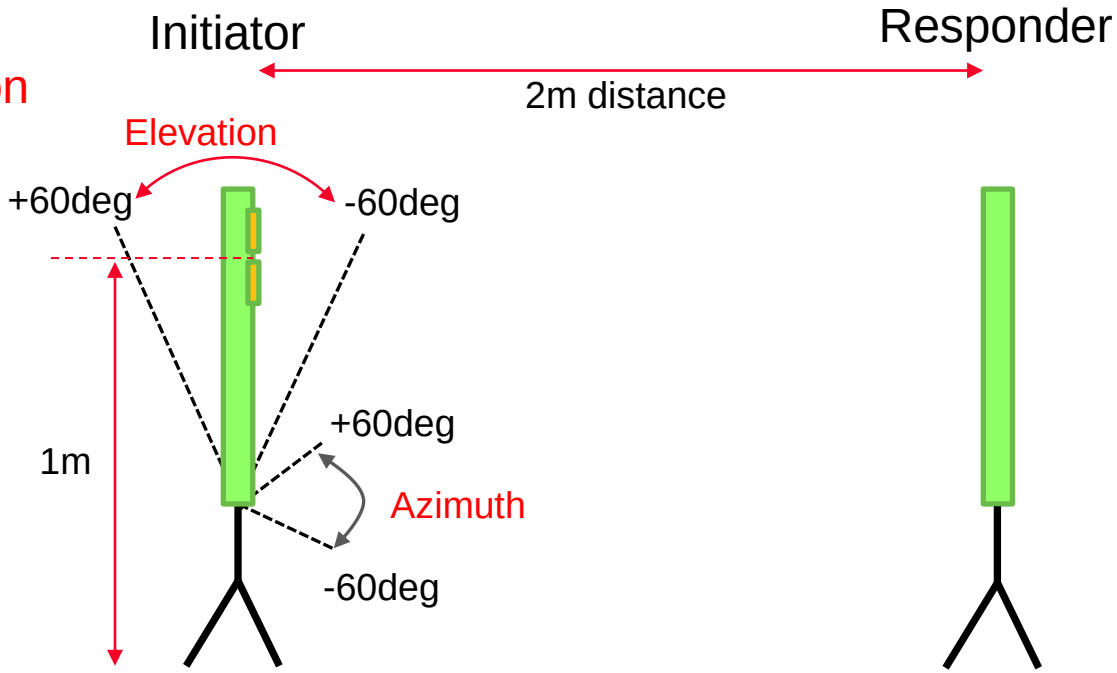
Measurement condition







Delay value

```
0xE4, 0x00, 0x08,
        0x05, 0x3B,     # DELAY_CALIBRATION_VALUE Channel 5
        0x10, 0x3B,     # DELAY_CALIBRATION_VALUE Channel 6
        0x10, 0x3B,     # DELAY_CALIBRATION_VALUE Channel 8
        0xE9, 0x3A,     # DELAY_CALIBRATION_VALUE Channel 9
```

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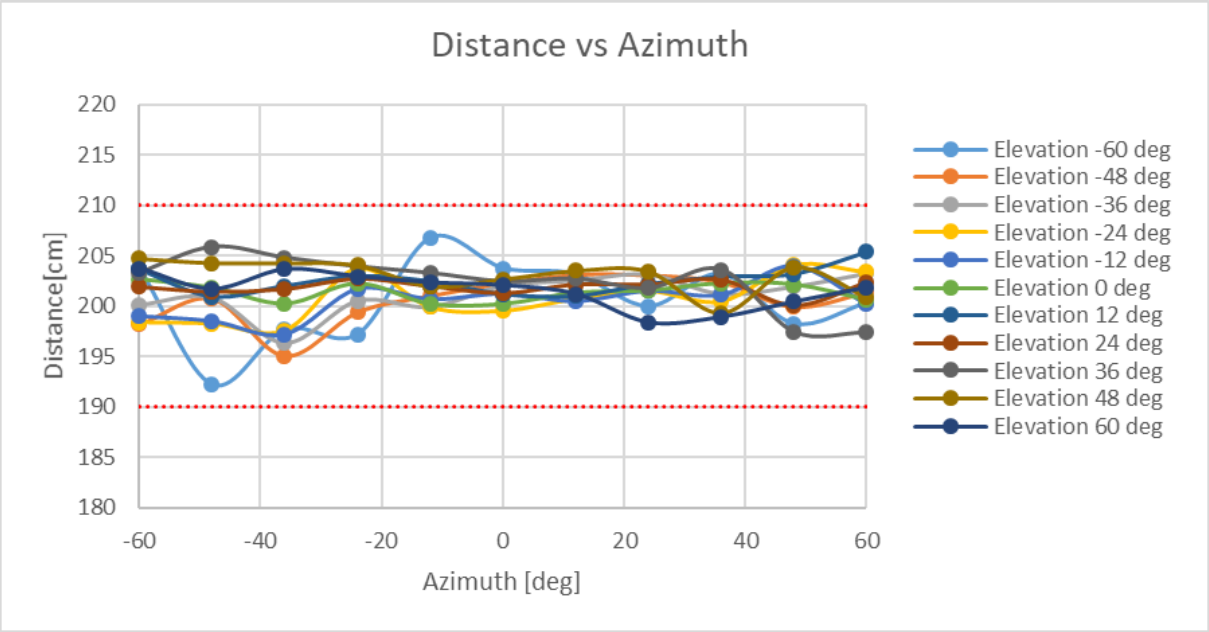
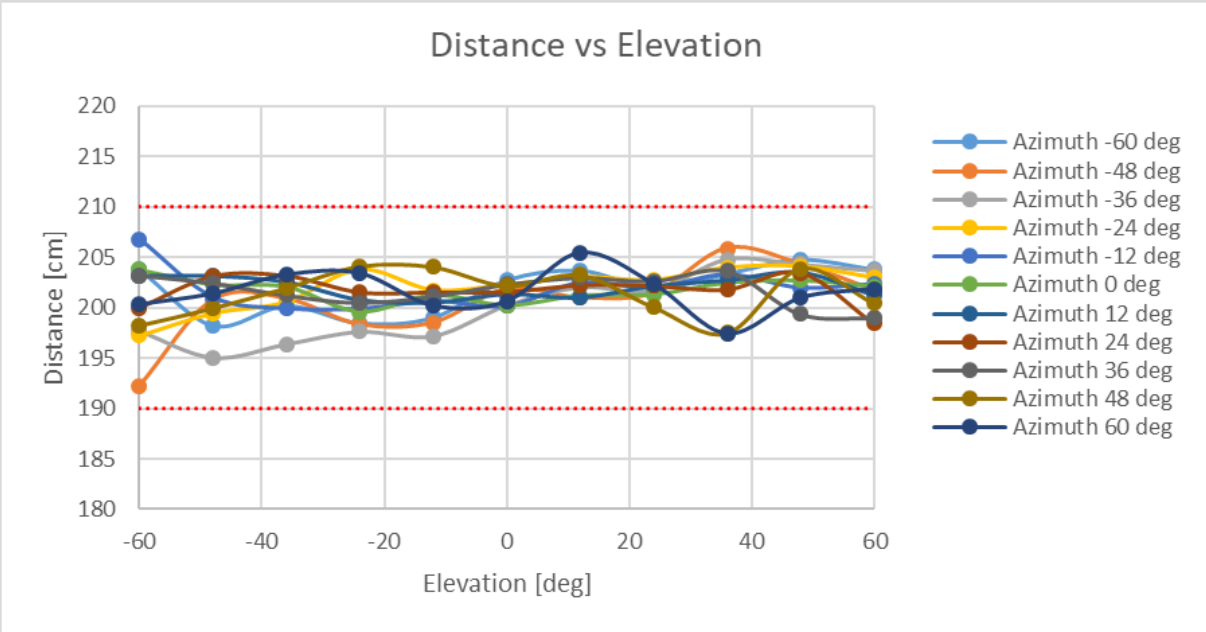
Distance (channel 9, DUT#B10)





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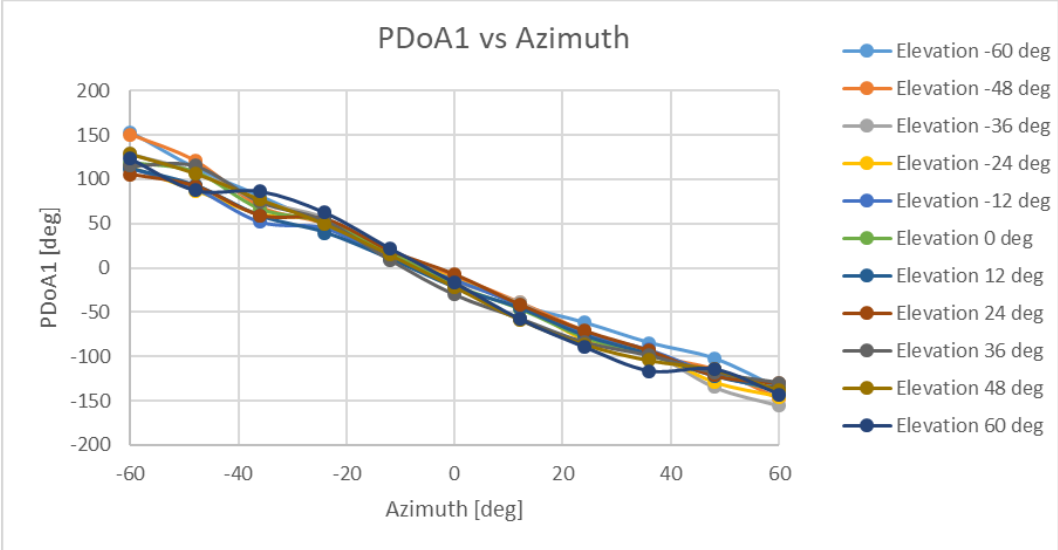
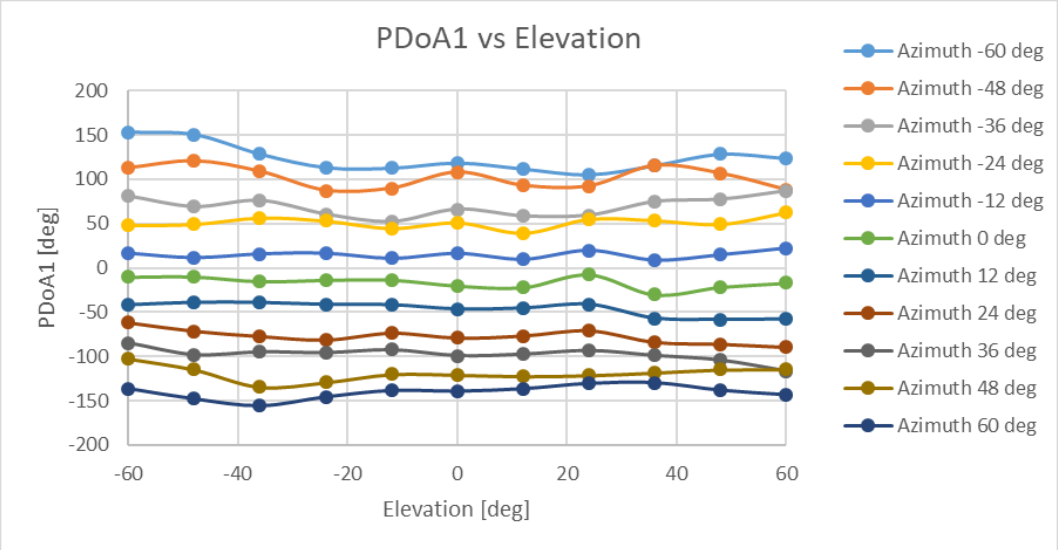
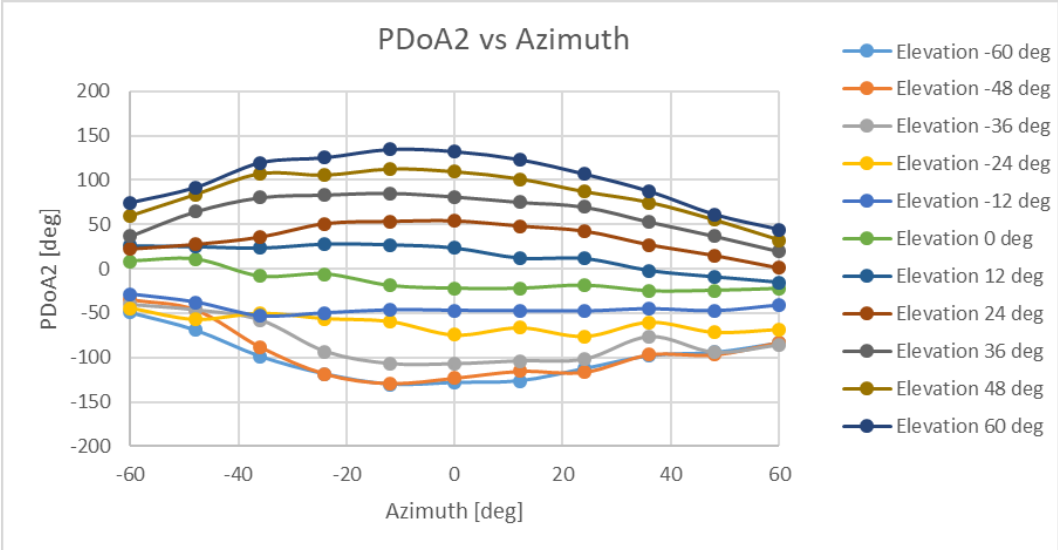
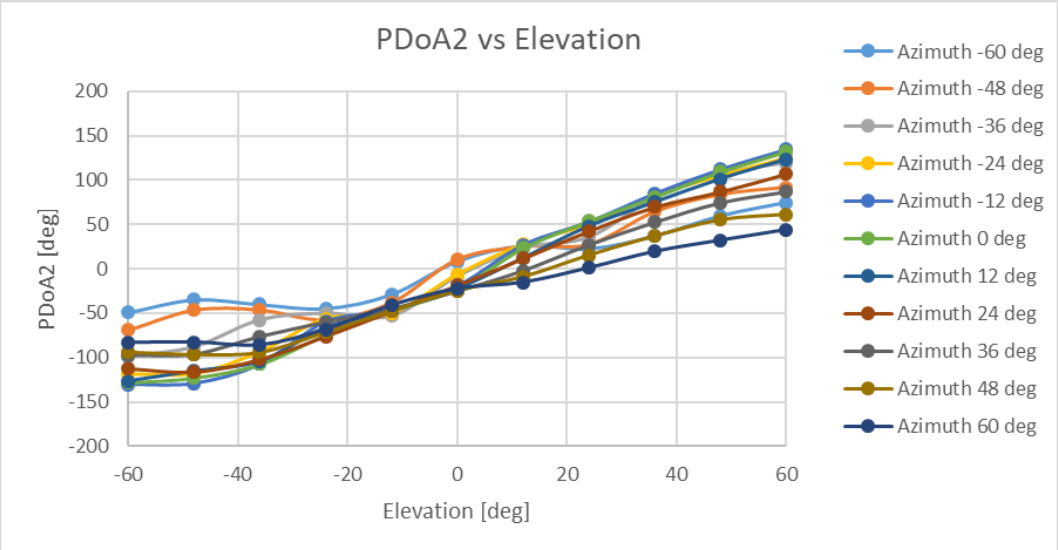
PDoA (channel 9, DUT#B10)









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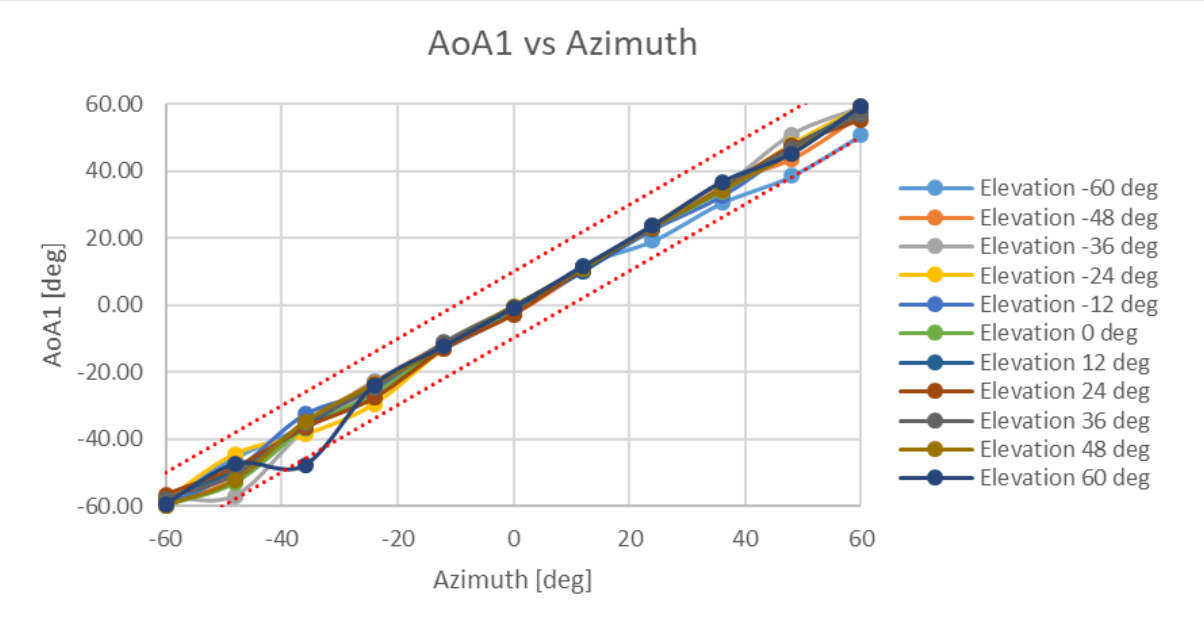
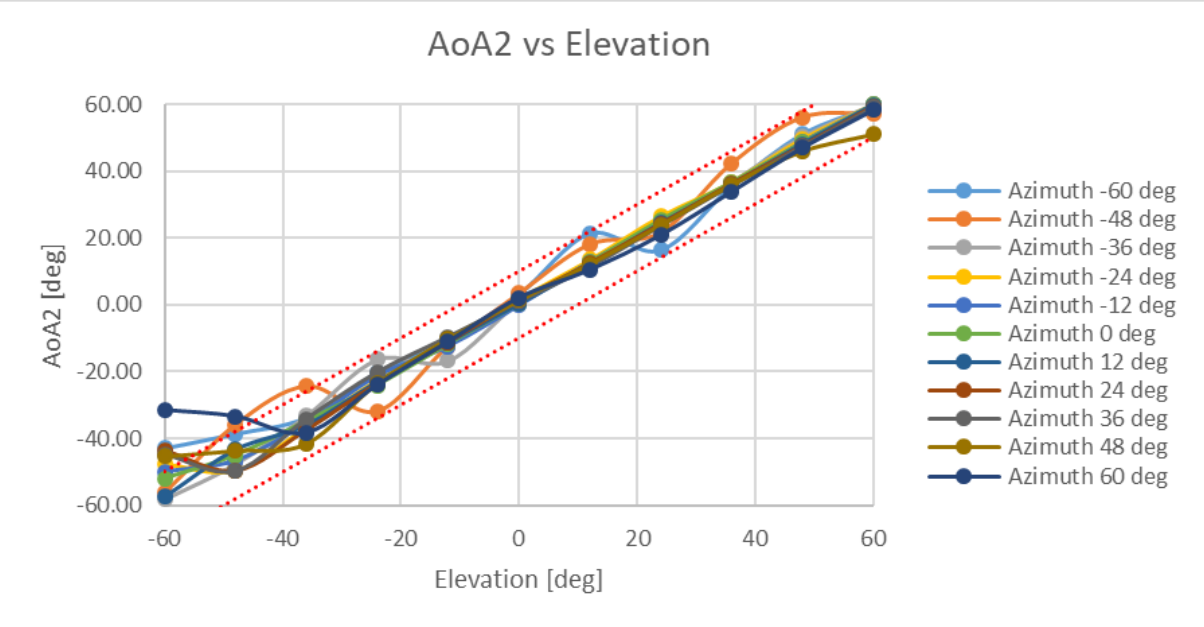
AoA (channel 9, DUT#B10)





	Elevation	Elevation	Elevation	Elevation	Elevation	Elevation	Elevation	Elevation	Elevation	Elevation	Elevation
	-60	-48	-36	-24	-12	0	12	24	36	48	60
Azimuth	deg	deg	deg	deg	deg	deg	deg	deg	deg	deg	deg
-60	-57.82	-59.48	-57.46	-58.21	-60.00	-59.98	-58.27	-56.57	-57.94	-59.94	-59.41
-48	-45.69	-51.96	-56.99	-44.44	-49.30	-53.18	-49.20	-48.49	-50.78	-52.32	-47.40
-36	-36.05	-35.56	-37.46	-38.71	-32.62	-36.99	-36.35	-36.63	-35.50	-34.98	-47.85
-24	-22.67	-23.05	-25.02	-29.57	-25.31	-26.62	-23.86	-27.74	-24.75	-23.72	-23.93
-12	-13.11	-11.73	-12.58	-13.07	-12.41	-12.55	-12.60	-13.13	-11.27	-12.46	-12.37
0	-2.28	-2.80	-1.31	-1.34	-1.87	-1.34	-1.04	-2.87	-0.54	-0.52	-0.88
12	11.72	10.25	10.98	10.67	10.20	11.13	10.00	10.91	11.31	10.83	11.44
24	19.06	23.81	23.84	23.07	22.44	22.93	23.14	23.00	23.11	23.26	23.49
36	30.45	36.25	34.95	34.64	32.53	33.81	34.42	34.77	34.73	34.21	36.62
48	38.47	43.53	50.85	48.02	46.39	46.09	47.32	47.34	46.72	45.12	45.03
60	50.71	56.64	59.02	59.08	59.07	58.61	57.36	55.36	56.94	59.25	59.25

	Azimuth	Azimuth	Azimuth	Azimuth	Azimuth	Azimuth	Azimuth	Azimuth	Azimuth	Azimuth	Azimuth
	-60	-48	-36	-24	-12	0	12	24	36	48	60
Elevation	deg	deg	deg	deg	deg	deg	deg	deg	deg	deg	deg
-60	-42.95	-56.20	-58.17	-47.66	-50.02	-52.06	-57.43	-43.61	-44.68	-45.27	-31.44
-48	-38.62	-35.77	-47.60	-49.64	-46.68	-45.08	-43.28	-49.67	-49.77	-43.57	-33.36
-36	-33.81	-24.23	-33.22	-36.74	-35.46	-34.82	-35.97	-37.45	-34.13	-41.49	-38.28
-24	-20.32	-31.75	-16.37	-22.56	-21.36	-24.09	-22.97	-23.55	-20.16	-23.49	-23.96
-12	-10.85	-12.36	-16.77	-11.44	-12.43	-11.77	-11.22	-11.21	-9.58	-10.21	-11.04
0	3.02	3.41	0.77	1.66	-0.23	0.32	0.28	1.29	1.05	1.39	2.14
12	21.28	18.18	12.53	13.45	12.85	12.70	11.90	12.38	11.40	12.19	10.50
24	16.35	21.69	24.02	26.43	24.82	25.41	24.38	24.07	23.71	23.82	20.88
36	35.38	42.33	36.69	36.71	36.07	36.66	35.93	36.16	35.73	35.89	33.87
48	50.96	56.17	50.26	49.64	48.78	48.64	48.10	47.49	47.37	45.91	46.97
60	59.41	57.34	59.93	59.95	59.89	59.89	59.69	59.31	59.12	50.98	58.29

in 5deg	in 10deg	over 10deg

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Revision history

Rev	date	Description
A	Nov 11th 2022	Initial release
B	Dec 26th 2022	Add 2D AoA calibration

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